

California Housing Price Prediction

Using Linear Regression

Artificial Intelligence & Machine Learning Internship
Short Project Report

Project	California Housing Price Prediction
Dataset	California Housing Dataset (Scikit-learn)
Model	Linear Regression
Train/Test Split	80% / 20%
MAE	0.533
RMSE	0.746
R² Score	0.576

1. Introduction

This project demonstrates a basic end-to-end Machine Learning workflow for predicting median house values using the California Housing dataset. The workflow covers data loading, exploratory data analysis (EDA), data preparation, train-test splitting, model training, evaluation, visualization, and model saving. The project was implemented in Python using Google Colab, Pandas, NumPy, Matplotlib, Joblib, and Scikit-learn.

2. Dataset Description

The California Housing dataset is available through Scikit-learn using `fetch_california_housing()`. It contains 20,640 observations and eight numerical input features describing housing districts in California. The target variable, **MedHouseVal**, represents the median house value.

The input variables include median income, house age, average rooms, average bedrooms, population, average household occupancy, latitude, and longitude. Because the features are numerical, categorical encoding is not required for this model.

3. Exploratory Data Analysis

EDA was performed before model training to understand the dataset structure and basic statistical characteristics. The first rows were inspected using `df.head()`, while `df.shape` was used to determine the dataset dimensions. `df.info()` provided information about data types and non-null entries, and `df.describe()` provided descriptive statistics such as mean, standard deviation, minimum, maximum, and quartiles.

Missing values were checked using `df.isnull().sum()`. The built-in California Housing dataset does not contain missing values, so no missing-value imputation was required. The project also uses an Actual vs Predicted scatter plot to visually inspect the quality of the model's predictions.

4. Data Preparation

The target column **MedHouseVal** was separated from the input features. The resulting feature matrix was stored in **X**, while the target was stored in **y**. The data was divided into training and testing sets using an 80:20 split with `random_state=42` to make the experiment reproducible.

Ordinary Linear Regression does not require feature scaling for this task, so a scaling step was not added. This keeps the preprocessing simple and appropriate for the selected baseline model.

5. Model Training

A Linear Regression model was selected as the baseline regression algorithm. Linear Regression estimates a linear relationship between the input features and the target variable by fitting coefficients that minimize prediction error on the training data.

The model was trained using the training portion of the dataset and then used to generate predictions for the unseen test data. The trained model was subsequently saved using Joblib as `linear_regression_model.pkl`, allowing it to be reused without retraining.

6. Model Evaluation

Three standard regression metrics were used: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 score. MAE measures the average absolute prediction error, while RMSE gives greater influence to larger errors. R^2 measures the proportion of variation in the target that is explained by the model.

Metric	Result	Interpretation
MAE	0.533	Average absolute prediction error
RMSE	0.746	Penalizes larger prediction errors more strongly
R^2	0.576	Model explains approximately 57.6% of target variation

The model achieved an MAE of **0.533**, an RMSE of **0.746**, and an R^2 score of **0.576**. Therefore, the model explains approximately 57.6% of the variation in the target variable on the test set. The RMSE being higher than the MAE suggests that some larger prediction errors are present.

The California Housing target is represented in units of \$100,000 in the Scikit-learn dataset. Consequently, an MAE of 0.533 corresponds to an average absolute error of approximately \$53,300, while an RMSE of 0.746 corresponds to approximately \$74,600.

7. Actual vs Predicted Analysis

An Actual vs Predicted scatter plot was generated after prediction. The horizontal axis represents the actual house values and the vertical axis represents the values predicted by the Linear Regression model. A diagonal reference line represents ideal predictions. Points closer to this line indicate smaller prediction errors, while points farther away indicate larger errors.

8. Improvement Ideas

- Feature engineering could create more informative variables from the existing housing features.
- Ridge and Lasso Regression could be evaluated as regularized alternatives to ordinary Linear Regression.
- Tree-based models such as Random Forest or Gradient Boosting could capture non-linear relationships.
- K-fold cross-validation could provide a more robust estimate of model performance.
- Hyperparameter tuning could be applied when using models with configurable parameters.
- Additional EDA, including correlation analysis and residual analysis, could provide deeper insight into model errors.

9. Conclusion

This project successfully implemented a complete introductory Machine Learning workflow using the California Housing dataset. The data was explored and prepared, a Linear Regression baseline was trained, and its performance was evaluated using MAE, RMSE, and R^2 . The resulting R^2 score of 0.576 indicates that the model captures a substantial portion of the variation in house values, while the error metrics show that prediction errors remain significant. The trained model was saved for future use. More advanced regression methods, feature engineering, and cross-validation could be investigated to improve predictive performance.